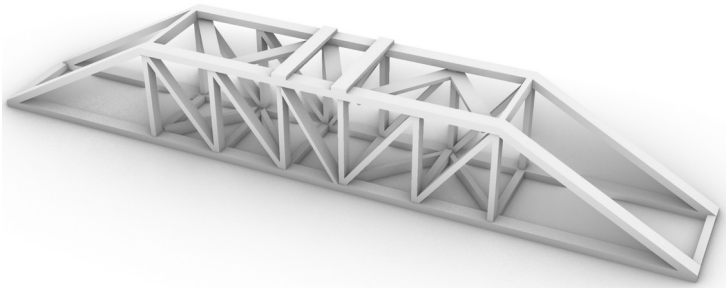
Introduction to Structural Design, Spring 2026
Professor Ochsendorf
Corey Segal and Busra Nimet Yesil
April 8, 2026

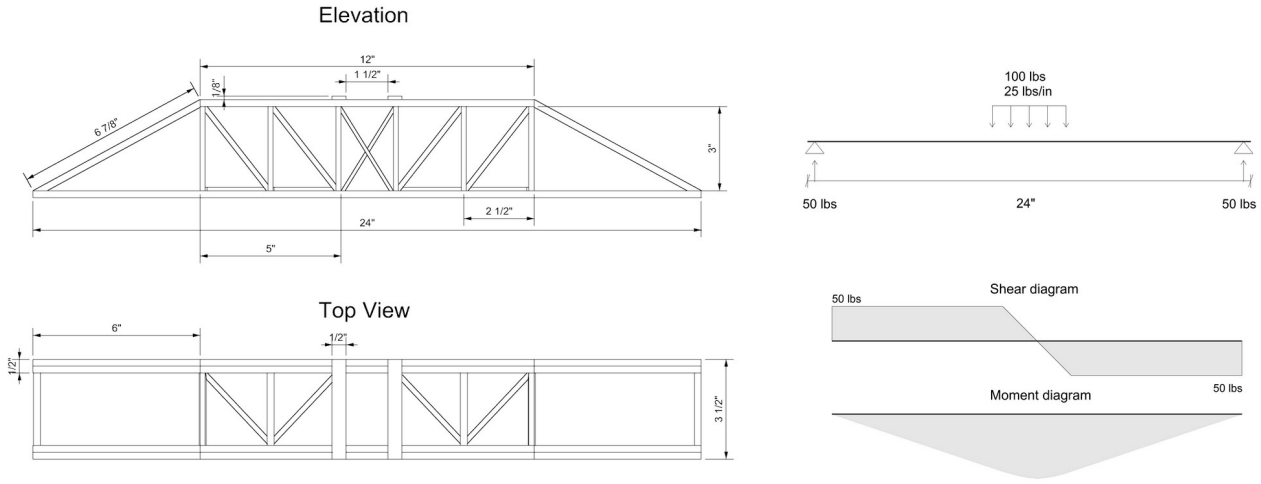
The “Economic” Beam





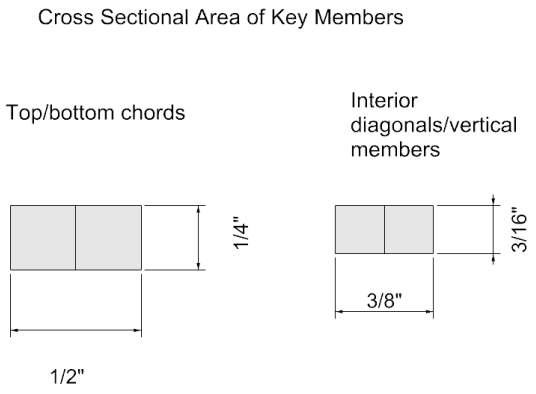
Description of Design

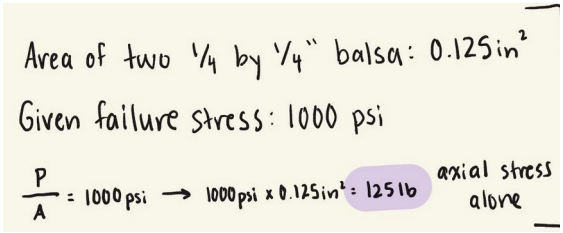
The beam was analyzed as a simply supported truss spanning 24” with a total design load of 120 lb (instead of 100lb for safety) applied over the 4” center loading region, as required in the project statement. The design consists primarily of two trapezoidal trusses that roughly followed the moment diagram and were connected with horizontal bracing.

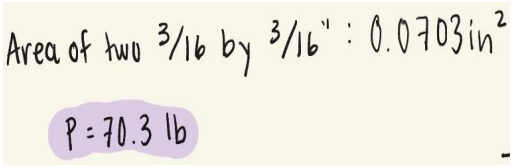


The final beam uses two identical side trusses in a 3D box configuration that helped prevent sideways buckling, so each side truss was assumed to carry one-half of the total load. Static equilibrium and truss force relationships showed that the largest axial forces occur near midspan, with the top chord and side diagonals in compression and the bottom chord in tension. Because axial forces are always higher near the midspan, the most common area of buckling in a beam is in the top chord. Due to this, we wanted the top chord to be thick.

Because these members were the ones primarily undergoing the most force, the top and bottom chords were built from two ¼” × ¼” sticks to make them more robust, giving a total area of 0.125 in² and an axial stress capacity of about 125 lb using the given allowable stress of 1000 psi: Failure Stress, fcr =F/A = 1000psi, therefore 1000 psi x 0.125 in² = 125 lb.The vertical members were built from two 3/16” × 3/16” sticks, giving an area of 0.0703 in² and an axial capacity of about 70 lb per member (1000 psi x 0.0703 in² = 70 lb).

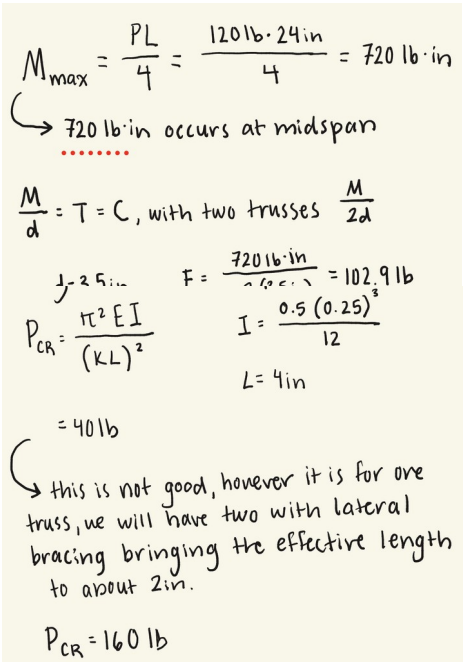






The vertical members and short interior diagonals were made from the previous lab’s column, making the structure economical through reuse/minimizing materials. Further, angling the diagonal members at each end as a result of following the moment diagram rather than having a box beam throughout further reduced material and improved the design’s efficiency.

These trussed diagonals were kept lighter (doubled 3/16 by 3/16”) because the Pratt-type orientation places them primarily in tension and they were not the primary members we expected to fail. The top chord was also checked using Euler buckling, showing that the 3D lateral bracing between the two side trusses is necessary to prevent premature sideways buckling. Based on these calculations, the beam was proportioned so that the center top chord is the most likely governing member and the predicted failure mode is buckling near midspan.



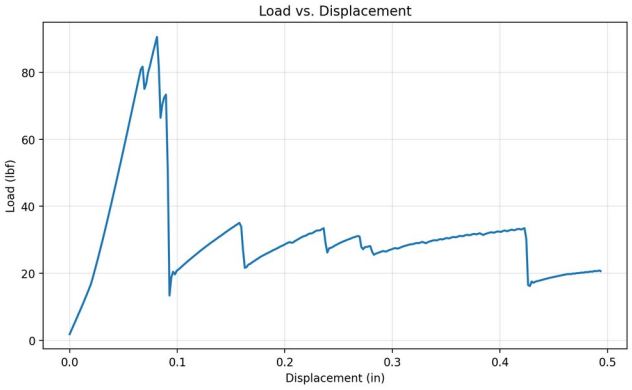
Initial Calculations

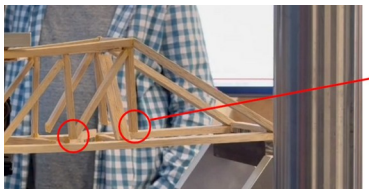
We sized the depth to be 3.5” given the limit was 4”. For the top and bottom chords in each truss, we calculated an axial force of 102.9 lbs. We rounded this to 100 lbs: the top chord carries 100 lbs of compression and the bottom chord carries 100 lbs of tension. The trussed diagonals carry around 50 lbs of tension and 2 center verticals carry 35 pounds of compression. From these conservative estimates, we had to determine the amount of balsa needed to withstand these forces in each member.

From these numbers, we built the top and bottom chords out of doubled ¼” x ¼” balsa wood and everything else with doubled 3/16” by 3/16” recycled balsa wood . From there, we conducted a buckling check for the top chord; this is where we expected the highest compressive force to be initially, and thus, our

final expected failure mode was buckling at the center top chord. This is why we additionally braced it without including those members in our calculations.

Load vs Displacement Curve





Failure Mode Justification

At failure, the right-end support reaction was approximately 45.3 lb-f based on static equilibrium of the beam at the peak recorded load of 90.56 lbf. An analysis of the right end panel joints show that this reaction produced a large axial force in the end diagonal, on the order of 79 lb-f, together with a horizontal force component of about 65 lb-f transferred into the adjacent joint (see below). This caused significant joint rotation and eccentric loading in the right end vertical members. Because these verticals were balsa members with subpar joints, they were vulnerable to instability and connection failure, especially if the wood glue joint wasn't holding properly i.e. didn't have enough surface area to stick on. This was exacerbated by the fact that we put the interior diagonals in tension, since tension relies more on the joints than compression does. The observed behavior, in which one vertical lifted off and another flew out, is therefore consistent with end-panel joint separation followed by progressive loss of stability, rather than simple axial crushing of a member. We believe that if it weren't for this joint, we could have reached 100 pounds. The large horizontal force from the diagonal to the right joint could cause rotation of the joint and peeling/failure of the glue which would explain why the vertical was lifted.

Left support = right support = P / 2
Force at right end joint = 90.56 / 2 = 45.28 lbf
Length of diagonal = 6.10 in

Diagonal force = 79 lbf
Horizontal force from diagonal ≈ 64.7 lbf
Bottom chord force into right end = 64.7 lbf